

Quick Check

Lesson 3: Seed Dormancy: Why Do Healthy-Looking Seeds Not Germinate?

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 3

Mr. Wafula planted Rosecoco beans and some healthy-looking seeds did not germinate even though the rain, soil and planting day were the same. Which statement best describes a dormant seed?

- A** A living seed that fails to germinate even when conditions are favourable
- B** A seed that has been eaten by pests in the soil
- C** A dead seed whose embryo has rotted
- D** A seed that fails because it was planted in waterlogged soil

Of 100 Rosecoco bean seeds planted, 20 did not germinate. The reasons were: 8 had a hard impermeable seed coat, 6 were in a waterlogged patch, 4 were old seed with dead embryos, and 2 were eaten by pests. How many of the 20 non-germinating beans failed for reasons OTHER than dormancy?

A 12

B 14

C 8

D 6

In Mr. Wafula's field, 12 maize seeds did not germinate: 5 were eaten or rotted by fungi or pests, 4 were old seeds with dead embryos, and 3 were in a waterlogged patch. Among the 20 non-germinating beans, 8 had a hard impermeable seed coat. Which conclusion is supported by this data?

- A** Maize seeds can never be dormant
- B** Dormancy caused all the bean failures but none of the maize failures
- C** Dormancy explained 8 of the 20 bean failures and none of the 12 maize failures in this field
- D** The hard seed coat caused most of the maize failures

The 8 hard-coated bean seeds were scarified (nicked) and replanted. By Day 7, 7 of the 8 had germinated. What percentage of the 8 seeds germinated?

A 70%

B 7.8%

C 87.5%

D 12.5%

Why did scarifying (nicking or rubbing) the hard seed coat of the dormant bean seeds help them germinate?

- A** It added food to the seed so the embryo could grow
- B** It made the embryo larger before planting
- C** It made a small opening in the impermeable testa so water and oxygen could reach the embryo
- D** It killed the pests living inside the seed

A farmer finds that some seeds stay dormant because of a chemical inhibitor, abscisic acid, in the seed. Which way of breaking dormancy matches this cause best?

- A** Planting the seeds in a waterlogged patch
- B** Soaking the seeds in water for some hours so the inhibitor is washed out
- C** Rubbing the seed coat with sandpaper
- D** Storing the seeds in sunlight and adding fertiliser

A learner says, 'A seed that does not germinate must be dead.' Which group of non-germinating bean seeds from Mr. Wafula's field shows that this idea is wrong?

- A** The 4 old seeds with dead embryos
- B** The 8 hard-coated seeds, of which 7 germinated after scarification
- C** The 5 maize seeds eaten or rotted by fungi
- D** The 2 seeds eaten by pests

Why can dormancy be useful to a wild plant but a problem for a farmer like Mr. Wafula?

- A** Dormancy helps seeds survive dry or cold seasons and germinate over time, but the farmer wants all seeds to germinate together for a uniform stand
- B** Dormant seeds always die, so wild plants make more seeds, but the farmer loses money
- C** Dormancy stops pests from eating seeds in the wild but not in a farm
- D** Dormancy makes seeds grow faster, which wild plants need but farmers do not

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).